

Instructor: Date:

Learner: Regional Branch:

Program: Center:

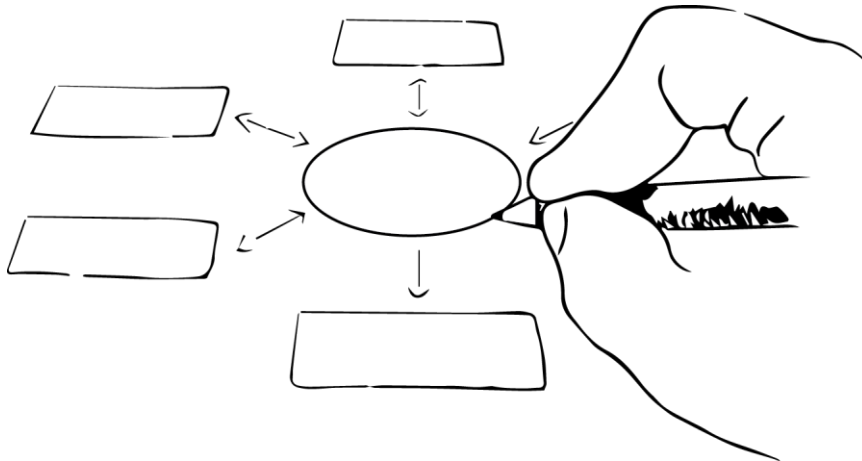
CONTENT

- Let's Start! • Let's Explore! • Let's Practice! • Extension Activity • Target Words • Learning Points



Let's Start!

I want to know more about component diagrams.



Learning Outcome:

Describe common activities at work to a visitor using short sentences in familiar spoken and written situations.
Give detailed information related to UML and common related topics.

Can Do Statements:

- I can explain how the component diagram works.
- I can give some guidelines related to component diagram.
- I can identify key features related to the component diagram.

Ask your partner:

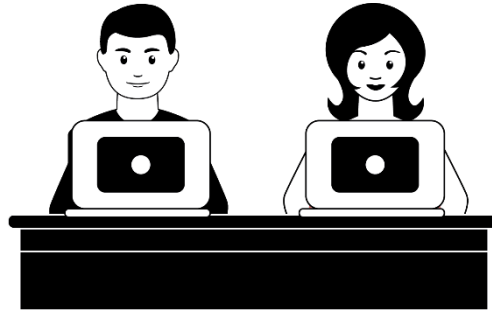
- When studying, do you divide information into smaller chunks? Why?
- Is it better to show how components of a system interact? Why? Why not?





Let's Explore!

We need to decide what diagram to use for this project.



Instructions: Read the text below.

- Arthur:** We've got to take a look at the different diagrams that we can use for our next model.
- Ennis:** I was thinking about the component diagram, but we've got to check if it's the most suitable. What do you think Amber?
- Arthur:** I know that component diagrams are an important part in model architecture because they show how components of a system connect to each other to create bigger components. When we start designing a model with these patterns, we'll have the foundation of the piece of software we want to develop.
- Ennis:** Ok. We've got to take into account how this diagram works. Let me give you an example, this diagram has components that we can use to execute a stereotype function. We can wire these components together when we use an assembly connector; we have got to link this connector to what we call a lollipop and a socket symbol. Now, remember we have got to model components as rectangles, and when we have the foundation of the system in this model, we will proceed to the high-level development stage. We just need to work hand-in-hand.
- Arthur:** That'd be both convenient and efficient for this project. But we also need to evaluate the limitations of component diagrams. I think that once we have the first stage or low-level development ready, we have got to find a way to deal with physical architecture or hardware issues, if necessary.
- Ennis:** But I think that in this stage we can use it, even though it is not widely used anymore. When we have the software model defined, we'll address hardware. The best we can do is start with the development of the basis for this software.
- Arthur:** You're right! This application might not require too much tweaking for it to work on different hardware environments or dependencies. We've got to focus on keeping cohesion between components, assign different interfaces to components, and what else?
- Ennis:** Hierarchies are quite important, we need to define different classes, how they'll interact and we have to minimize the size of messages to assure a good flow between objects.

**Let's Practice!****Component diagram process**

Instructions: Underline the correct word to complete the sentence.

1. First, coworkers (has got / got / have got) to create the outline of the diagram.
2. Then, they've got (for connecting / to connect / connect) the components to the other parts if needed.
3. They have got to (use / using / used) the lollipop and socket symbol.

**Let's Practice!**

When you start using the component diagram, you will need to know several things about it.

Instructions: Select the correct answer to complete the sentence.

1. When we _____ designing a model with these patterns, we'll have the foundation of the piece of software we want to develop.
 - a. start
 - b. starting
 - c. starts
2. When we use the component diagram, we _____ have to model components as rectangles.
 - a. ill
 - b. will
 - c. wills
3. When we create the foundation of the system using this model, we will _____ to the high-level development stage.
 - a. proceed
 - b. proceeds
 - c. proceeded
4. When we connect the components together, we _____ an assembly connector.
 - a. won't use
 - b. will use
 - c. will used
5. When we are ready to use the components, we _____ a stereotype function.
 - a. want execute
 - b. will execute
 - c. executing



Extension Activity

Component diagram manual

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for information and examples of how component diagram works.
- Write a guideline for component diagram.
- Identify key features of component diagram.
- Collect the information and design a manual that helps you to understand component diagram process.
- Present the manual to your instructor and partners

Self-Assessment

Yes

No

- I collect useful information about component diagrams.
- I understand how component diagrams work.
- I design a manual to help me understand how the component diagram works.
- I explain key features and guidelines related to component diagrams.

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Target Words

- find a way** (expression): to solve, to find a solution or to disentangle something
e.g. we have got to find a way to deal with physical architecture or hardware issues, if necessary.
- tweaking** (noun): change, alteration or fixing
e.g. This application might not require too much tweaking for it to work on different hardware environments or dependencies.
- cohesion** (noun): well-connected, structured and logically organized
e.g. We've got to focus on keeping cohesion between components, assign different interfaces to components, and what else?
- suitable** (adjective): appropriate, adequate or acceptable
e.g. I was thinking about the component diagram, but we've got to check if it's the most suitable.
- pattern** (noun): a model, design or guide to do something
e.g. When we start designing a model with these patterns, we'll have the foundation of the piece of software we want to develop.
- foundation** (noun): the basis or elemental structure of something
e.g. When we start designing a model with these patterns, we'll have the foundation of the piece of software we want to develop.



Learning Points

Future Time Clauses With When

You use the future tense to talk about distant future or things that you haven't planned.

In order to use the future tense, you can follow this structure:
subject + will + main verb (base form) + complement.

Example:

*Component diagrams **will** become the most important way to develop business software.*

If you want to make negative sentences you can use the following structure:
subject + won't/will not + main verb (base form) + complement.

Example:

*Diagram icons **won't** disappear, but they'll adapt to developers' needs.*

You use **future time clauses with when** to refer to something that will happen as a result of a possible action. In this case, you are joining a dependent clause to an independent one. You usually write the *dependent clause (when)* in **present** , and the *independent clause* in **future** .

Example:

*I'll start using that diagram **when** we reach an agreement.*

You can alternate the position of the clauses by separating them using a comma:

Example:

***When** the project is finished, we won't discard the design.*

Examples

- *When we have the foundation of the system in this model, we will then proceed to the high-level development stage, and even interfaces.*
- *When we have the software model defined, we'll address hardware.*



Learning Points

Have Got

You use **Have Got** to talk about obligations, necessity or possession.

In order to use Have got, you need to take into account that the verb *Have* changes depending on the subject it follows, take a look at the following chart:

Subject	Have
I	Have
You	Have
He	Has
She	Has
It	Has
We	Have
They	Have

You can use Have Got to express obligation or necessity by following the next structure:

subject + have got to /has got to + main verb (base form) + complement

Examples:

We have got to use component diagrams in this project.

She has got to come early tomorrow because she didn't finish creating that diagram.

Examples

- *We've got to take a look at the different diagrams that we can use for our next model.*
- *We've got to focus on keeping cohesion between components, assign different interfaces to components*